

Chronic bicipital tenosynovitis in dogs: 29 cases (1985–1992)

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Summary: Medical records of 23 dogs with unilateral and 3 dogs with bilateral chronic bicipital tenosynovitis were reviewed. Mean age of affected dogs was 4.6 years (SD, 2.0 years), and mean body weight was 32.6 kg (SD, 14.5 kg). Neither a breed nor a gender predilection was detected. All dogs had a history of intermittent or progressive weight-bearing lameness that became worse after exercise. Mean duration of lameness prior to medical or surgical treatment was 6.5 months (range, 0.25 to 24 months). In all dogs, signs of pain were evident during palpation of the biceps tendon within the intertubercular groove. Radiography revealed sclerosis or osteophytosis of the intertubercular groove in all 29 shoulder joints. Mild degenerative joint disease was evident radiographically in 17. Arthrography was performed in 12 joints, and in 11 there were irregularities of or filling defects along the biceps tendon. Arthrocentesis was performed on 17 joints; 14 synovial fluid samples had cytologic abnormalities consistent with degenerative joint disease.

Medical treatment, consisting of injection of methylprednisolone acetate into the biceps tendon and its synovial sheath, was attempted in 21 of the 29 affected shoulder joints. Surgery, which consisted of tenodesis of the biceps tendon, was attempted in 14 joints; 8 of these had not been treated medically; the remaining 6 had poor results following medical treatment.

Gross and histologic findings consistent with chronic bicipital tenosynovitis were observed in all 14 joints in which surgery was performed. Seventeen of the medically treated shoulders were available for clinical evaluation, and results were excellent or good in 7. Twelve of the surgically treated shoulders were available for clinical re-evaluation, and results were excellent or good in all 12 (mean duration of follow-up, 5.7 months; range, 2 to 13 months). Owners of all dogs were contacted by telephone. Owners reported that results were excellent or good in 10 of the 21 medically treated shoulder joints, and in all 14 of the surgically treated shoulder joints (mean duration of follow-up, 30.1 months; range, 4 to 82 months).

Complications developed in 3 of the 4 dogs in which an osteotomy of the greater tubercle had been performed (implant migration, 2 dogs; delayed union, 1 dog). A seroma developed in 1 of the 10 dogs in which tenodesis was performed by laterally transposing the biceps tendon through a hole in the greater tubercle. Complications related to medical treatment were not detected.

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Bicipital tenosynovitis is an inflammation of the biceps brachii tendon and its surrounding synovial sheath. The condition reportedly is a common cause of forelimb lameness in mature dogs,¹ and clinically is evident as a chronic, mild to marked, often debilitating weight-bearing lameness.

Although bicipital tenosynovitis has been described in dogs,^{1-5,a,b} much of the information currently available regarding its clinical features and the long-term response of affected dogs following treatment is speculative or anecdotal.^{1-3,5,a,b} For instance, authors of 1 report described clinical signs in 6 dogs with bicipital tenosynovitis,⁴ but did not evaluate long-term efficacy of treatment.

Recommended treatments for bicipital tenosynovitis include injection of repository corticosteroids into the tendon and surgical transposition of the biceps brachii tendon.^{1-5,a,b} However, there apparently have not been any studies of the long-term efficacy of treatment. Therefore, the purposes of the study reported here were to identify the clinical, radiographic, and pathologic features of bicipital tenosynovitis in dogs, and to evaluate long-term results of medical and surgical treatment of affected dogs.

Criteria for Selection of Cases

Medical records of all dogs in which bicipital tenosynovitis was diagnosed between January 1985 and December 1992 were reviewed. Dogs that underwent medical treatment alone were included in the study only if they had physical examination findings (eg, signs of pain during palpation of the biceps tendon in the intertubercular groove) and radiographic signs (eg, sclerosis or osteophytosis of the intertubercular groove and mineralization of the biceps tendon) considered typical of bicipital tenosynovitis.¹ Dogs that underwent surgery were included in the study only if the diagnosis had been confirmed on the basis of gross and histologic examination of affected tissues. In addition, all dogs were included in the study only if their owners were available to provide follow-up information. Dogs were excluded from the study if they had any other orthopedic or neurologic disease in the same limb as the limb with bicipital tenosynovitis.

Procedures

Signalment, body weight, activity level, affected shoulder joint, duration of lameness, history, physical examination findings, and results of radiography, positive-contrast arthrography, and synovial fluid analysis for each dog were recorded. Efficacy of treatment was determined by evaluating clinical information recorded at the time of each dog's last physical examination (eg, gait, limb function, range of motion in the affected joint, and

whether signs of pain were evident during manipulation of the affected joint or palpation of the affected tendon). Results were considered to be excellent (normal gait, limb function, and range of motion, no signs of pain during palpation of affected tendon), good (normal gait, limb function, and range of motion, signs of mild pain during palpation of the affected tendon or manipulation of the affected joint), fair (mild lameness, but improved compared with condition prior to surgery, signs of pain on palpation of the affected tendon and manipulation of the affected joint), or poor (marked lameness that was more severe than before surgery, poor range of motion in affected joint, and signs of pain during palpation of affected tendon and manipulation of affected joint). Results were also assessed on the basis of information obtained during follow-up telephone interviews with the dogs' owners. Owners were asked questions regarding their dogs' activity level, degree of lameness, and ability to return to work, and were asked whether the dogs were lame following activity or inclement weather, or needed anti-inflammatory medications. Results were considered to be excellent (dog is never lame), good (dog is lame only after heavy or strenuous activity), fair (dog is lame after moderate or normal activity), or poor (dog is always lame and did not improve or was worse after treatment).

The surgical procedure performed and time to maximal limb usage following surgery also were recorded, as were gross and histopathologic changes seen in affected tissues. Complications resulting from surgical or medical treatment were recorded.

Statistical Analysis

Age, breed, and gender of all affected dogs and of all other dogs admitted to the veterinary teaching hospital during the same 7-year period were compared by means of Pearson's χ^2 test⁶ (breed, gender) or Student's *t*-test (age). In addition, Pearson's χ^2 test was used to compare breed, gender, and activity level of affected dogs that underwent medical treatment along with breed, gender, and activity level of affected dogs that underwent surgery. Student's *t*-test was used to compare age, weight, and pretreatment lameness duration for dogs that underwent medical treatment alone with age, weight, and pretreatment lameness duration for dogs that underwent surgery. Logistic regression⁷ and log-linear models^{8,9} were used to test the relationship between outcome (excellent, good, fair, or poor) and the following variables: medical or surgical management, pretreatment activity level, pretreatment lameness duration, age, weight, dose of repository corticosteroids, and number of corticosteroid injections. The relationships between surgeon and outcome and between surgical procedure and outcome also were analyzed by use of a log-linear model. For all analyses, a value of $P < 0.05$ was considered significant. A computer-based statistics program^c was used to perform all analyses.

Results

Signalment—Of 83 dogs in which bicipital tenosynovitis had been tentatively diagnosed, 23 dogs with unilateral bicipital tenosynovitis and 3 dogs with bilateral bicipital tenosynovitis met the criteria for inclusion in

the study. In the 3 dogs with bilateral bicipital tenosynovitis, the shoulder joints were affected at different times.

The 26 dogs that were included in the study were between 1.2 and 9.6 years old (mean, 4.6 years; SD, 2.0 years; median, 4.5 years). Most were middle-aged or older; 77% were more than 3 years old. Seven of the dogs were Labrador Retrievers or Labrador Retriever-type dogs, 2 were Rottweilers, 2 were German Shepherd Dogs, 2 were Samoyeds, and 1 was a mixed-breed dog. The remaining 11 dogs represented 11 different breeds. Thirteen dogs were male, and 13 were female. Dogs weighed between 11.4 and 71.4 kg (mean, 32.6 kg; SD, 14.5 kg; median, 30.2 kg). Only 15% of the dogs weighed less than 20 kg. Sixteen of the dogs were athletic or sporting dogs; the remaining 10 were obese, sedentary dogs. In the 23 dogs with unilateral bicipital tenosynovitis, the right shoulder joint was affected in 16 and the left was affected in 7.

History—All dogs had a history of intermittent or progressive, weight-bearing forelimb lameness that became worse after exercise. In 25 of the 26 dogs, the lameness was chronic. Duration of lameness prior to examination at the veterinary teaching hospital ranged from 0.25 to 24 months (mean, 6.5 months; median, 5.5 months). Prior to evaluation at the veterinary teaching hospital, 19 dogs had been treated with 1 or more of the following: aspirin (14 dogs), prednisone (9 dogs), phenylbutazone (1 dog), polysulfated glycosaminoglycan^d (2 dogs), meclofenamic acid^e (1 dog), or tetracycline and penicillin (1 dog that was suspected to have borreliosis). In all dogs, the lameness was refractory.

Physical examination findings—In all dogs, signs of pain were evident during palpation of the biceps tendon within the intertubercular groove and during manipulation (ie, extension) of the shoulder joint. Atrophy of the supraspinatus and infraspinatus muscles was observed in 4 of the 29 affected joints.

Results of diagnostic tests—Radiography revealed sclerosis or osteophytosis of the intertubercular groove in all 29 shoulder joints (Fig 1). In 17 joints, osteophytes could be seen on the caudal aspect of the humeral head or the caudal aspect of the glenoid rim. These osteophytes were interpreted as evidence of mild degenerative joint disease. Mineralization of the biceps tendon was observed in 6 joints. Arthrography was performed on 12 joints, and in 11 of the 12, there were irregularities of or filling defects along the biceps tendon (Fig 2).

Arthrocentesis was performed on 17 joints. In 14, the concentrations of monocytes, macrophages, and vacuolated phagocytes in the synovial fluid were high, which was considered indicative of degenerative joint disease.¹⁰ Total nucleated cell counts from the synovial fluid samples ranged from 1,500 to 4,800 cells/ μ l (mean, 2,785.7 cells/ μ l; median, 1,900 cells/ μ l). Mucin content of the synovial fluid was low in 1 joint. In all other joints examined, mucin content and viscosity of the synovial fluid was normal.

Medical treatment—Twenty-one of the 29 affected shoulder joints were treated medically. Treatment consisted of injecting methylprednisolone acetate^f (10 to 40

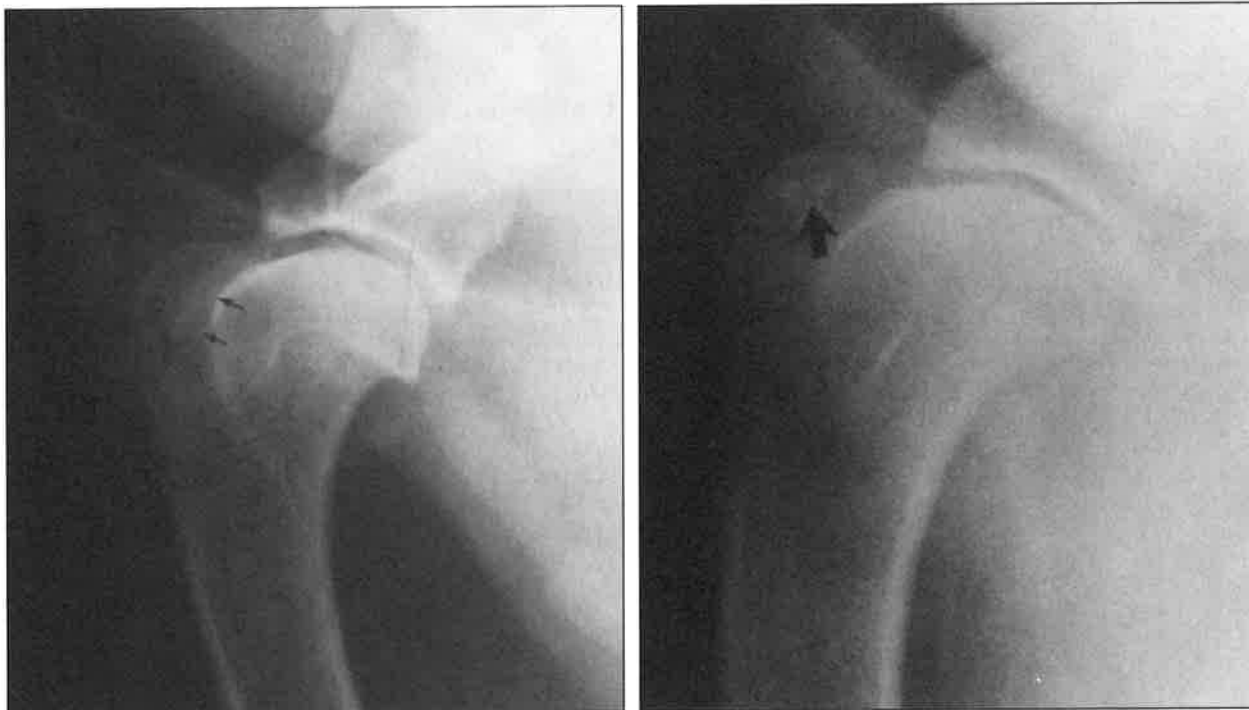


Figure 1—Mediolateral radiographic views of the shoulder joints of 2 dogs with chronic bicipital tenosynovitis. (Left) Osteophytes can be seen in the intertubercular groove (arrows). (Right) Focal mineralization of the biceps tendon is evident (arrow).

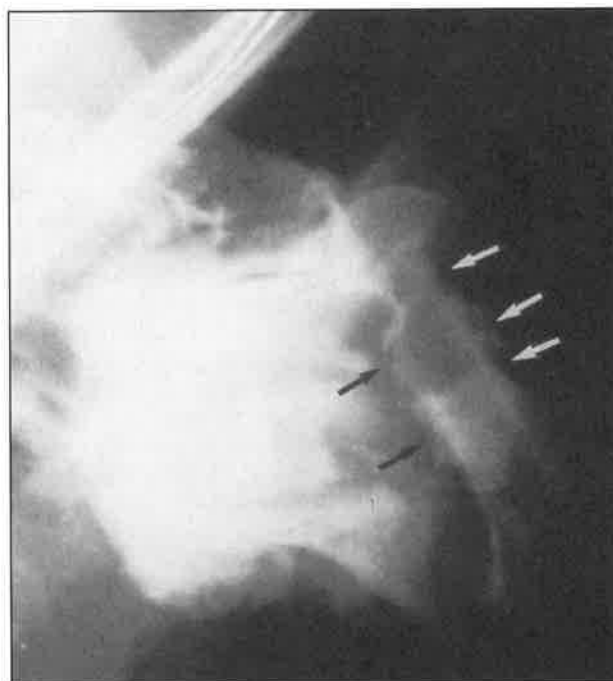


Figure 2—Positive-contrast arthrogram of the shoulder joint of a dog with chronic bicipital tenosynovitis. Notice the filling defects and irregularities along the biceps tendon (arrows).

mg) into the tendon and tendon sheath and restricting exercise for 2 weeks. If the lameness did not improve, prednisolone injections were repeated at 2-week intervals.

The number of injections given per shoulder ranged from 1 to 3. It was not possible to determine from the medical records how well owners complied with the sug-

gestion that exercise be limited for 2 weeks after each injection.

Seventeen of 21 medically treated shoulders (16 dogs) were available for in-hospital reevaluation. Based on clinical assessment, results of medical treatment were considered excellent in 3, good in 4, fair in 4, and poor in the remaining 6 (follow-up time ranged from 1 to 36 months; mean, 5.6 months). During telephone interviews, owners reported that the results of medical treatment were excellent for 3 shoulder joints, good for 7, fair for 5, and poor for the remaining 6 (mean duration of follow-up, 23.6 months; range, 4 to 82 months). There was no statistical relationship between outcome of medical treatment and age, body weight, lameness duration, activity level, number of treatments, or treatment dose. There were no complications associated with medical treatments. All 6 dogs with poor response to medical treatment subsequently underwent surgery.

Surgical treatment—Surgery, which consisted of tenodesis of the biceps brachii tendon, was performed on 14 shoulder joints. In 8 of these, medical treatment had not been attempted; in the remaining 6, results following medical treatment were poor.

A cranial approach to the shoulder joint¹¹ was used. In 10 joints, tenodesis of the biceps brachii tendon was performed by laterally transposing the tendon through a hole drilled in the greater tubercle.¹ In the remaining 4 joints, tenodesis was performed by laterally transposing the tendon following osteotomy of the greater tubercle.¹ The greater tubercle was replaced in each dog by placing 2 K-wires and a figure-8 tension band wire. Surgical procedures were selected on the basis of the personal preference of the surgeon. In 6 instances, a Velpau sling was applied postoperatively and maintained for approx-

imately 10 days. Exercise was restricted postoperatively in all dogs for 6 to 8 weeks.

Twelve of 14 shoulders (11 dogs) were available for in-hospital reevaluation. Based on clinical assessment, results of surgery were considered excellent in 8, and good in 4 (follow-up time ranged from 2 to 13 months; mean, 5.7 months). None of the dogs had impaired shoulder function or was lame. Two dogs had evidence of mild pain during manipulation of the shoulder joint. During telephone interviews, owners reported that the results of surgery were excellent for 11 shoulder joints, and good for 3 (mean duration of follow-up, 28.1 months; range, 4 to 62 months). All owners reported a marked improvement in their dogs' condition following surgery, and reported that their dogs were able to return to a level of activity similar to the level the dogs had had prior to the onset of the tenosynovitis. There was no statistical relationship between outcome of surgical treatment and age, body weight, lameness duration, activity level, prior medical treatment, surgeon, or surgical procedure performed. There was no significant difference between dogs treated medically and those treated surgically in regard to signalment, activity level, body weight, or lameness duration.

Complications of surgical treatment—Complications developed in 3 of the 4 dogs in which an osteotomy of the greater tubercle was performed. One dog had delayed union of the greater tubercle. The osteotomy did eventually heal, and outcome in this dog was excellent. In the other 2 dogs, migration of a K-wire resulted in lameness. The implants were removed, and outcome was good in 1 dog and excellent in the other.

The only complication reported in the 10 shoulders in which tenodesis was performed by passing the tendon through a hole in the greater tubercle was formation of a seroma in 1 dog. Limb function was not adversely affected by the seroma. Warm compresses were applied to the affected area, and the seroma resolved quickly. One dog did develop a temporary contracture of the elbow joint, but this was considered to be a result of placing the limb in a Velpeau sling.

Time to achieve good or excellent clinical results following surgery ranged from 2 to 9 months (mean, 3.5 months; median, 3 months). In 10 dogs, maximal limb usage was reported within 2 to 3 months after surgery, but in the remaining 4, recovery was prolonged and ranged from 4 to 9 months. In 2 of these dogs, the prolonged recovery was a result of migration of an implant. In the other 2 dogs, the reason for the delayed recovery was not determined.

Pathologic findings—Gross abnormalities visible in the 14 surgically treated shoulder joints included mineralization of the biceps tendon ($n = 2$), osteophytosis of the intertubercular groove ($n = 3$), proliferative synovitis ($n = 8$), and fibrous adhesions between the biceps tendon and its sheath ($n = 9$). Biopsy specimens were obtained from all 14 joints, and all had histologic evidence of proliferative tenosynovitis characterized by synovial proliferation, edema, fibrosis, and lymphocytic-plasmacytic infiltration of the tendon and synovium. Dystrophic mineralization of the biceps tendon was seen

histologically in 2 of 14 joints, and cartilaginous metaplasia of the tendon was seen in 2 others.

Discussion

The biceps brachii muscle originates from the supraglenoid tubercle of the scapula by means of a long tendon that crosses the shoulder joint and continues distally along the humerus through the intertubercular groove. The tendon is held in the groove by the transverse humeral ligament and invaginates the joint capsule of the shoulder, so that the joint capsule surrounds the tendon, forming a synovial sheath.^{4,12} It is this intracapsular, extrasynovial region of the tendon that is affected with bicipital tenosynovitis.

Bicipital tenosynovitis has been reported in human beings,¹³⁻¹⁶ dogs,^{1-5,a,b} and horses.¹⁷ As in this study, most affected dogs are middle-aged to older, and of medium or large size. However, there does not appear to be any gender or breed predilection. In human beings, isolated bicipital tenosynovitis is rare, and components of the rotator cuff are usually involved.¹⁵ In dogs, although degenerative joint disease is commonly found in conjunction with bicipital tenosynovitis, other ligaments and tendons of the shoulder joint do not appear to be involved.

In dogs, bicipital tenosynovitis has been reported secondary to acute trauma, tendon strain or rupture,⁴ and entrapment of joint mice within the tendon sheath.^{4,18} A previous traumatic injury to the affected shoulder joint was reported in 3 of the 26 dogs in this study, and it is possible that this injury contributed to the development of bicipital tenosynovitis in these dogs. In the remaining 23 dogs, however, an inciting cause was not identified, and the bicipital tenosynovitis was considered primary.

The cause of primary bicipital tenosynovitis remains unknown. Various authors have hypothesized that it is related to activities that cause overuse and chronic repetitive trauma,^{1-2,13} and propose that high biomechanical stress and repetitive trauma can damage the tendon, causing an inflammatory response.^{15,19} Although this theory seems plausible, dogs in this study comprised 2 distinct groups in regard to activity level. Sixteen dogs were active and athletic (eg, 5 of the 7 Labrador Retrievers and Labrador Retriever-type dogs were hunting or field trial dogs); but the remaining 10 were overweight, sedentary, and poorly conditioned. Strenuous activities performed by the 16 athletic dogs could have been the cause of bicipital tenosynovitis. However, why the 10 sedentary dogs developed this condition, and why dogs in poor physical condition in general seem to be predisposed to this condition,¹ is less clear.

In this study of 29 episodes of bicipital tenosynovitis, the right shoulder joint was affected (19/29, 66%) almost twice as often as the left shoulder joint (10/29, 34%). The reason for this is not known. In people, the dominant side is generally affected (ie, the right shoulder joint is more likely to be affected than the left in people who are right handed).¹³ It is possible that dogs possess a dominant side as well. In horses, bicipital bursitis has been reported to develop secondary to abnormal gaits caused by navicular disease.²⁰ All 3 dogs with bilateral bicipital tenosynovitis had hip dysplasia, and in 2 of the 3, hip dysplasia was of sufficient severity to require total

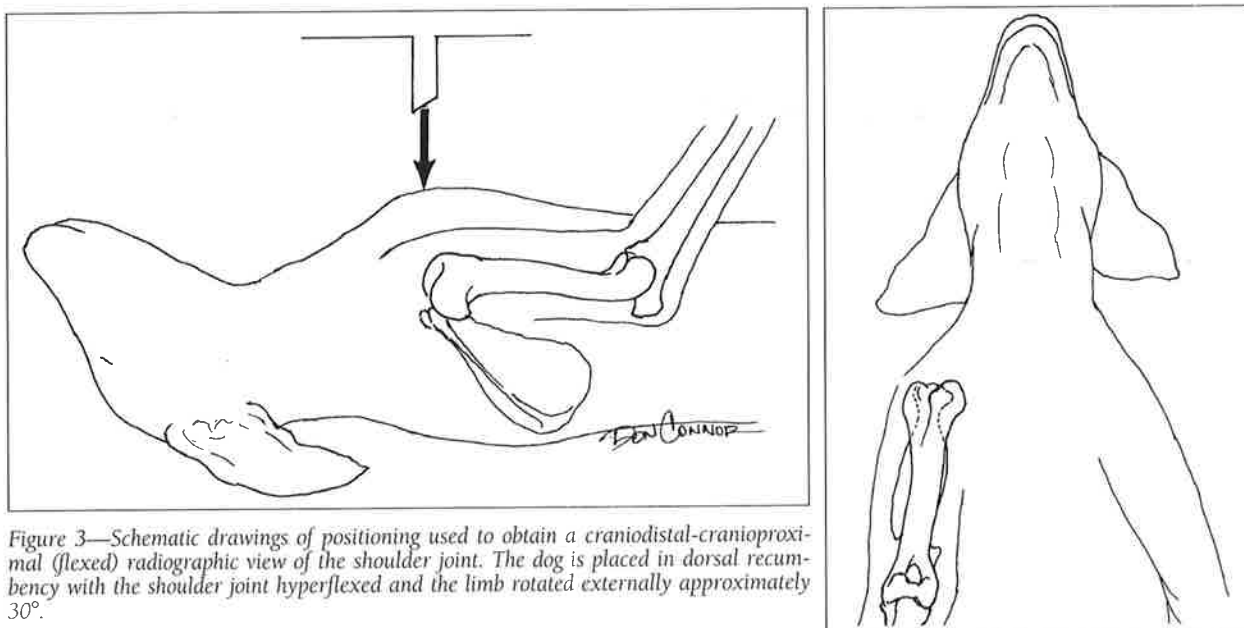


Figure 3—Schematic drawings of positioning used to obtain a craniodistal-cranioproximal (flexed) radiographic view of the shoulder joint. The dog is placed in dorsal recumbency with the shoulder joint hyperflexed and the limb rotated externally approximately 30°.

hip replacement. It is possible that hip dysplasia caused the dogs to shift their weight forward, predisposing them to bicipital tenosynovitis.

Bicipital tenosynovitis is often difficult and frustrating to diagnose in dogs because clinical and radiographic abnormalities can be subtle. However, a combination of historical, physical examination, radiographic, and clinicopathologic findings will often allow other disorders of the shoulder joint to be excluded. In reviewing medical records for this study, we identified several dogs in which bicipital tenosynovitis was initially diagnosed on the basis of physical examination findings that were subsequently found to have some other disease. Final diagnoses in these dogs included osteosarcoma of the proximal portion of the humerus ($n = 1$), brachial plexus neoplasia ($n = 2$), osteosarcoma of the cervical vertebrae ($n = 1$), cervical intervertebral disk disease with nerve root compression ($n = 3$), polyarthritis ($n = 2$), and sarcoma of the axillary region with lysis of the second sternbra ($n = 1$). Similar findings have been reported by others.¹ Because dogs were included in this study only if they had radiographic changes consistent with bicipital tenosynovitis, it is possible that some dogs with acute bicipital tenosynovitis were excluded because there had not been sufficient time for radiographic changes to develop. What effect this may have had on the results is unclear.

All dogs in this study had a weight-bearing forelimb lameness that became worse after exercise. In some, the lameness was chronic and progressive, but in others it was intermittent. Because pain occurs only when the tendon is gliding along the bone, dogs are not reluctant to bear weight on the affected limb. However, affected dogs often guard against flexion and extension of the shoulder, which limits the swing phase of the gait.^{1,4} Some authors have reported that the lameness can be exacerbated by flexing the shoulder joint for 1 minute.²¹ The lameness appears to be refractory to oral administration of anti-inflammatory drugs and did not resolve in any of the 19 dogs that were treated in this manner.

Results of palpation of the biceps tendon in the intertubercular groove during flexion and extension of the shoulder joint were highly suggestive of bicipital tenosynovitis, and signs of pain could consistently be elicited in all dogs in this study. Surprisingly, we did not detect any relationship between lameness duration and severity of atrophy of the infraspinatus and supraspinatus muscles, suggesting that atrophy may be more related to the severity of the disease than to its duration.

Because radiographic changes can be subtle even in chronic cases, we recommend that a craniodistal-cranioproximal (flexed) radiographic view of the intertubercular groove be obtained in addition to standard caudocranial and mediolateral views. To obtain this view, the dog should be positioned in dorsal recumbency, with the affected shoulder joint hyperflexed and the limb rotated externally approximately 30° (Fig 3). In this view the intertubercular groove is isolated, making it easy to identify subtle irregularities in the groove and mineralization of the biceps tendon. These changes may not be detected on the standard caudocranial and mediolateral views, because of superimposition of the intertubercular groove and proximal end of the humerus⁸ (Fig 4).

Contrast arthrography is useful for identifying abnormalities associated with bicipital tenosynovitis in dogs^{1-3,18,22,23} and human beings.^{14,15} Filling defects along the tendon correspond to proliferative synovium, adhesions between the tendon sheath and tendon, or joint mice.^{1,18,22} In this study, contrast arthrography was a reliable diagnostic tool; however, the high percentage of positive results (11/12) was likely related to the chronic nature of the cases selected for study.

Ultrasonography has been used to diagnose bicipital tenosynovitis and other disorders of the biceps tendon in horses¹⁷ and human beings,²⁴ and is considered by some to be superior to contrast arthrography in human beings.²⁴ However, results of a recent prospective study suggest that ultrasonography is not as sensitive as arthrography for diagnosing bicipital tenosynovitis in dogs.²²

The synovial sheath surrounding the biceps tendon is continuous with the shoulder joint capsule, and not surprisingly, most (14 of the 17) of the shoulders examined had synovial fluid abnormalities consistent with degenerative joint disease.¹⁰ Cytologic examination of synovial fluid appeared to be a useful ancillary diagnostic test in these dogs; however, it is not known why synovial fluid samples from 3 of 17 joints were normal in this study.

All dogs in this study that underwent surgery had gross and microscopic lesions indicative of chronic inflammation, and similar findings have been reported by others.^{4,13,16} In people, repeated stress or injury to the tendon results in hemorrhage and edema of the tendon and its surrounding synovium, which incites an inflammatory response and leads to cellular infiltration and formation of adhesions between the tendon and the synovial sheath.^{13,16} As the disease progresses, synovial proliferation and tendon fibrosis develop, and dense fibrous adhesions form between the tendon and its sheath.^{13,16} Disturbances in the blood supply to the tendon lead to tissue hypoxia, which in turn triggers transformation of the tendon into fibrocartilage. The fibrocartilage subsequently undergoes chondrocyte-mediated osteogenesis, similar to endochondral ossification.²¹

Excellent or good results were achieved in only 10 of the 21 medically treated joints in this study. This poor response was probably related to the chronic nature of the disease in these dogs. Further studies are needed to determine if dogs with acute bicipital tenosynovitis are more likely to respond favorably to medical treatment.

Dogs that did respond to medical treatment did so after 1 or 2 doses of methylprednisolone acetate. Surprisingly, neither the dose of corticosteroid nor the number of injections was significantly associated with outcome. Investigators have demonstrated that corticosteroids can impair the biomechanical properties and metabolic activities of articular cartilage,^{25,26} and infiltration of tendons with corticosteroids has been associated with collagen necrosis and tendon rupture.²⁷ Therefore, we suggest that prolonged medical treatment should be avoided in dogs with bicipital tenosynovitis, and that surgery should be considered for dogs that do not improve after 1 or 2 doses of corticosteroids.

The goal of surgery in dogs with bicipital tenosynovitis is to eliminate pain caused by interaction of the biceps tendon with the inflamed tendon sheath and osteophytes in the intertubercular groove. In this study, excellent or good results were achieved in all 14 of the dogs that were surgically treated, and all performance dogs were able to return to work. Long-term success rates of 90 to 94% following surgery have been reported in people as well.^{13,16}

Surgical transposition of the biceps tendon does not alter shoulder joint stability or function of the biceps tendon.²⁸⁻³⁰ Although tenodesis can induce mild, nonprogressive, degenerative changes in the shoulder joint,³¹ these changes do not appear to be clinically relevant.³² It has been suggested that dogs treated surgically early in the course of the disease process respond better than those treated later.¹ However, this was not the case in the present study. We did not detect a significant relationship between duration of lameness and the outcome of medical or surgical treatment.



Figure 4—Craniodistal-cranioproximal (flexed) radiographic view of the shoulder joint in a dog with chronic bicipital tenosynovitis. Notice how this view highlights areas of mineralization within the intertubercular groove (arrow).

Although outcome was satisfactory in all dogs that underwent surgery, time to full return to function was prolonged in some dogs. In people, mean time to full recovery following biceps tenodesis has been reported to be 8 months,¹³ but a small number of people have recovered times of 12 months or greater.^{13,16}

The rating scale used to evaluate outcome in these dogs was similar to that used by others.³³ Because outcome was based largely on owner evaluation, it was necessarily somewhat subjective. However, our results do suggest that a prospective study of treatment of bicipital tenosynovitis that includes a more objective measurement of lameness, such as force plate analysis,^{34,35} is warranted.

Complications associated with surgery in these dogs were few and relatively minor. Although the number of cases was small, our results suggest that lateral transposition of the tendon through a hole in the greater tubercle may be preferred over an osteotomy of the greater tubercle, because implants do not have to be used. Ultimately, however, both procedures resulted in good to excellent limb function. External coaptation following surgery did not appear to be necessary and may possibly have caused joint contracture in 1 dog.

^aHohn RB, Harrison JW. The pectoral limb: bicipital tenosynovitis, in *Proceedings*. 42nd Ann Meet Am Anim Hosp Assoc 1975;428.

^bPiermattei DL. Orthopedic conditions of the shoulder region, in *Proceedings*. 47th Ann Meet Am Anim Hosp Assoc 1980;371.

^cPROC GLM, SAS Institute Inc, Cary, NC.

^dAdequan, Luitpold Pharmaceuticals Inc, Shirley, NY.

^eArquel, Aveco Inc, Fort Dodge, Iowa.

^fDepo-Medrol, The Upjohn Co, Kalamazoo, Mich.

^gWeichselbaum R, College of Veterinary Medicine, University of Minnesota, St Paul, Minn: Personal communication, 1993.

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